

PYTHON PROGRAMMING

Course code	Course Title	No. Of Periods/ week	Total No of Periods/ Year	FA Marks	SA Marks	Credits
26AM105T	PYTHON PROGRAMMING	6	180	30	70	6

TIME SCHEDULE

Chapter No.	Chapter/UnitTitle	No. of Periods	Marks	No.of Short Answer Questions	No. of Essay Type Questions	CO's Mapped
1.	Introduction to Python Programming	40	14	02	01	CO1
2.	Control Flow and Loop statements	25	14	02	01	CO2
3.	Functions and Arrays	40	25	03	02	CO3
4.	Built-in Data Structures	35	22	02	02	CO4
5.	Python Libraries, File Handling and Exception handling.	40	25	03	02	CO5
Total		180	100	12	08	

COURSE OBJECTIVES

Upon completion of the course, the student shall be able

- (i) To know the fundamentals of Python programming and to understand syntax, scripting and program execution in 'Python'
- (ii) To develop various python programs using control structures, functions & arrays
- (iii) To develop programs using data structures
- (iv) To Implement Python Libraries
- (v) To implement file handling & exception handling concepts

COURSE OUTCOMES

CO1	AM105.1	Explain Basic concepts of Python programming as well as scripting, debugging and execution.
CO2	AM105.2	Develop Python programs using Control statements & Loops.
CO3	AM105.3	Develop Python programs using Functions and arrays.
CO4	AM105.4	Develop Python programs using Built-in Data structures.
CO5	AM105.5	Develop Python application programs using Python libraries, exceptions and Files,

LEARNING OUTCOMES

1.0 Introduction to python programming

- 1.1 Recite history of Python.
- 1.2 List Python features
- 1.3 List and Explain Applications of Python
- 1.4 Explain Python Integrated Development and Learning Environment (IDLE)
- 1.5 Explain process of Running Python Scripts.
- 1.6 Explain Identifiers, Keywords, Indentation & Variables
- 1.7 List and explain various data types
- 1.8 Explain declaration& initialization of variables.
- 1.9 Explain Input and Output statements.
- 1.10 Explain formatted input output.
- 1.11 State the usage of comments
- 1.12 List and explain various Operators.
- 1.13 Explain Boolean values.
- 1.14 Explain Operator precedence rules.
- 1.15 State the purpose of modules.
- 1.16 Define functions.
- 1.17 List types of functions
- 1.18 List and Explain Built-in Functions.
- 1.19 Explain the Steps in Developing a simple python program and execution.

2.0 Control Flow and Loop Statements

- 2.1 List and explain various Control Flow constructs.
 - 2.1.1 if
 - 2.1.2 if - else
 - 2.1.3 if – elif - else
- 2.2 List and explain various Loop Statements.
 - 2.2.1 For Loop
 - 2.2.2 while loop
- 2.3 Use break, continue & pass statements.
- 2.4 Implement the python programs using control structures

3.0 Functions and Arrays

- 3.1 Give introduction to functions
- 3.2 Use Function Arguments: Default arguments, Keyword arguments, Variable Length arguments, Anonymous Functions & Return Statement.
- 3.3 List and explain Scope of variables
- 3.4 Explain creation of modules.
- 3.5 Explain importing of modules.
- 3.6 Explain PythonVariables:Namespace and scoping
- 3.7 Explain Python Packages
- 3.8 List and Explain Strings: String slices, immutability
- 3.9 List and Explain String functions and methods.
- 3.10 Explain about String module.
- 3.11 Explain about Python Arrays.
- 3.12 Explain accessing of elements in an Array.
- 3.13 Explain Array methods.

4.0 Built-in DataStructures

- 4.1 Explain Python Lists.
- 4.2 Describe Basic List Operations.
- 4.3 Explain List Slices.
- 4.4 Explain List methods.
- 4.5 Explain List reversing.
- 4.6 Explain mutability.
- 4.7 Explain aliasing.
- 4.8 Explain Cloning of lists.
- 4.9 Explain List comprehension.
- 4.10 Tuples.
 - 4.10.1 Explain Tuple assignment.
 - 4.10.2 Explain Tuple as returnvalue.
 - 4.10.3 Explain Tuple Comprehension
- 4.11 Dictionaries
 - 4.11.1 Explain creation of dictionary.
 - 4.11.2 Explain Dictionary operations and methods.
 - 4.11.3 Explain Dictionary Comprehension.
- 4.12 Explain Sets.

5.0 Python Libraries, File Handling& Exception Handling

- 5.1 Give introduction to python library.
- 5.2 Import modules in a python program
- 5.3 Use Python standard library's functions and modules
- 5.4 Turtle, Random, Datetime, CSV, Numpy& Pandas
- 5.5 Create Python library
- 5.6 Give introduction to Files
- 5.7 Explain opening and closing Files
- 5.8 Explain reading and writing Files
- 5.9 Use Standard Input, Output and error streams
- 5.10 Explain exception handling
 - 5.10.1 Define Error and Exception
 - 5.10.2 State the difference between Error and Exception
 - 5.10.3 Use user-defined exceptions.
 - 5.10.4 Use try except block.
 - 5.10.5 Explain raising exceptions.

CO-PO/PSO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	1	2	1	1	1	2	2	1	
CO2	3	2	2	2	1	1	1	2	2	2
CO3	3	2	2	1	1	1		2	2	2
CO4	3	1	2	1	1	1	3	2	2	2
CO5	3	1	2	1	2	3	2	2	2	2
Average	3	1.4	2	1.2	1.2	1.4	2	2	1.8	2

3=stronglymapped,2=moderatelymapped,1=slightlymapped

COURSE CONTENT

UNIT-I: Introduction

Introduction to Python, installation, Applications, IDLE, Data types, variables, Input and Output statements, Operators, expressions, statements, precedence of operators, comments, modules, functions, types of functions, built-in functions, flow of python program execution.

UNIT-II: Control Flow and Loop Statements:

Control Flow - if, if-else, if – elif - else, for, while, break, continue, pass statements.s ample python programs using control flow and Loop statements.

UNIT-III: Functions and Arrays:

Introduction to Functions: Defining, Calling Functions, Passing Arguments, Types of function arguments -Keyword Arguments, Default Arguments, Variable-length arguments, Anonymous Functions, FunctionReturning multiple values, Scope of the Variables in a Function – Global and Local Variables.

Modules: Creating modules, import statement, name spacing, Python Packages.

Strings: string slices, immutability, string functions and methods, string module.

Python arrays: Access the Elements of an Array, array methods.

UNIT-IV: Data Structures

Lists: list operations, list slices, list methods, list traversing, mutability, aliasing, cloning lists, list comprehension.

Tuples: tuple assignment, tuple as return value, tuple comprehension.

Dictionaries: operations and methods, comprehension.
sets.

UNIT – V: Python Libraries, File Handling and Exception Handling.

Python Libraries: Introduction, importing modules in python program, use python standard library functions and modules - Python libraries - Turtle, Random, Datetime, CSV, Numpy& Pandas.

File Handling: Open Files, File Processing and Closing a File.

Error and Exceptions: Difference between an error and Exception,

Handling Exception, try except block, Raising Exceptions, User Defined Exceptions

REFERENCES

1. Python Programing by K. Nageswara Rao, Shaikh Akbar - Scitech Publications (India) Pvt.Ltd.
2. Python Programming:AModernApproach,VamsiKurama,Pearson
3. Learning Python, MarkLutz,Orielly
4. Think Python,Allen Downey,GreenTea Press
5. CorePythonProgramming,W.Chun, Pearson.
6. Introduction to Python, Kenneth A.Lambert, Cengage

TABLE SPECIFYING THE SCOPE OF SYLLABUS TO BE COVERED
FOR UNIT TESTS

Unit Test	Learning outcomes to be covered
Unit test-1	From 1.1 to 2.1
Unit test-2	From 2.2 to 3.13
Unit test-3	From 4.1 to 5.15